AI OUTBOUND SALES & EMAIL AUTOMATION OS

Product Requirements Document (PRD)

Version 1.1 | 11 August 2026 | Email Warmup module added

Master product and engineering specification.

1. Executive Summary

The AI Outbound Sales & Email Automation OS is a self-hosted, AI-powered sales platform for agencies and B2B teams. " "It combines lead discovery, company/contact research, lead qualification, email verification, personalized outreach, " "multi-step email automation, deliverability controls, reply intelligence, CRM workflows, and campaign optimization in one system.

2. Product Vision

Enable an agency to move from ICP definition to qualified prospect, personalized outreach, intelligent reply handling, " "and sales pipeline management without depending on a collection of separate sales tools.

3. Core Product Pillars

- Personalization — research prospects and generate factual, relevant messaging.
- Email Automation — schedule and execute multi-step sequences with conditional follow-ups.
- Workflow Efficiency — automate repetitive prospecting, research, verification, outreach, and CRM updates.
- Deliverability — support authentication checks, suppression, bounce handling, sending controls, and health monitoring.
- Data-Driven Optimization — analyze campaign outcomes and recommend improvements.
- Reply Management — detect, classify, prioritize, and assist with replies using AI.

4. Goals and Success Criteria

- Provide a usable all-in-one outbound workflow from lead discovery/import to CRM.
- Support an initial target of approximately 100 emails/day as a configurable campaign limit, subject to mailbox/provider rules and applicable law.
- Reduce manual prospect research and personalization work.
- Prevent follow-ups after replies, hard bounces, or unsubscribes.
- Provide measurable campaign and funnel analytics.
- Keep external lead/data providers optional rather than architectural dependencies.
- Run the application on the existing Hostinger VPS with PostgreSQL, Redis, workers, and Docker.

5. Non-Goals / Explicit Constraints

- No dependency on Apollo, Instantly, Smartlead, GoHighLevel, Google Maps/Places, Yelp, Clutch, LinkedIn data APIs, Foursquare, Yellow Pages, or similar proprietary lead databases for the core product.
- No promise or guarantee of inbox placement; deliverability is influenced by mailbox/provider reputation and external factors.
- No collection of private/personal data through unauthorized means. Discovery must use permitted/public business information, user-provided datasets, or authorized APIs/connectors.
- Do not build an enormous proprietary global contact database as a prerequisite for V1.
- Do not implement deceptive, fabricated, or hallucinated personalization.

6. Target Users

- AI/software agencies selling automation and development services.
- Freelancers scaling into agencies.
- B2B sales teams.
- Founders doing outbound sales.

7. Primary User Journey

Define ICP → Discover/import leads → Enrich company → Discover business contacts → Verify email → AI qualify → Research → Personalize → Add to campaign → Schedule/send → Detect reply/bounce/unsubscribe → AI classify → Stop or continue workflow → CRM → Meeting/deal → Analyze results → Optimize next campaign.

8. Functional Requirements & Module Backlog

M01 — Authentication & Admin

01.01 Login — Secure login with email/password, session creation, validation, protected routes.

Acceptance: Auth page; secure session; invalid credentials rejected.

01.02 Logout — Invalidate session and return user to login.

Acceptance: Session no longer grants access.

01.03 Account Settings — Manage profile, timezone, agency information, and password.

Acceptance: Changes persist and are validated.

M02 — Dashboard

02.01 Overview — Show leads, qualified leads, verified emails, sent, replies, meetings, deals, revenue.

Acceptance: Metrics are calculated from system data.

02.02 Campaign Performance — Show sent, delivered, bounce, reply, positive-reply, meeting, and conversion metrics.

Acceptance: Charts/tables filter by date/campaign.

02.03 Today's Activity — Show scheduled emails, replies, follow-ups, meetings, and tasks.

Acceptance: Activity list is current.

M03 — ICP Builder

03.01 Create ICP — Define country, location, industry, size, revenue, technology, keywords, and target criteria.

Acceptance: ICP can be saved/reused.

03.02 Decision Makers — Define target titles such as Founder, CEO, COO, CTO, Head of Sales; allow custom titles.

Acceptance: Search/qualification uses titles.

03.03 Pain Points — Define business problems the offer solves.

Acceptance: Pain points are available to AI qualification/personalization.

03.04 Offer Definition — Define services/offers and value propositions.

Acceptance: Campaign AI can reference approved offers.

M04 — Lead Generation

04.01 Lead Search — Search using ICP criteria and show progress/results.

Acceptance: Search creates a background job and results.

04.02 Search Job — Create asynchronous search jobs and show progress.

Acceptance: UI remains responsive for large jobs.

04.03 Company Discovery — Discover permitted public business information.

Acceptance: Company records are normalized.

04.04 Normalization — Standardize names/domains/locations.

Acceptance: Equivalent records map consistently.

04.05 Deduplication — Prevent duplicate companies/leads using domain/name/contact identifiers.

Acceptance: Duplicate records are merged or skipped.

M05 — Website Intelligence

05.01 Website Crawler — Fetch permitted public company pages.

Acceptance: Respect robots/technical restrictions and rate limits.

05.02 Data Extraction — Extract description, services, products, locations, public contacts, technologies/signals.

Acceptance: Structured research record is stored.

05.03 Technology Detection — Identify publicly observable technology signals where appropriate.

Acceptance: Signals include source and timestamp.

05.04 Research Record — Persist research summary, signals, and timestamp.

Acceptance: AI can consume the research.

M06 — Contact Discovery

06.01 Decision Maker Identification — Prioritize relevant business titles.

Acceptance: Contact candidates receive source/confidence metadata.

06.02 Contact Record — Store name, title, company, business email if available, public profile URL, source, confidence.

Acceptance: Data is traceable to its source.

06.03 Contact Deduplication — Avoid duplicate contacts.

Acceptance: Unique constraints/merge logic work.

M07 — Email Discovery

07.01 Public Business Email Extraction — Extract publicly listed business emails from permitted sources.

Acceptance: Email source is recorded.

07.02 User Data Import — Support CSV/XLSX/manual data for existing leads.

Acceptance: Mapping and validation are available.

07.03 Email Confidence — Assign confidence before verification.

Acceptance: Confidence is visible and stored.

M08 — Email Verification

08.01 Syntax Validation — Validate email format.

Acceptance: Invalid syntax rejected.

08.02 Domain Validation — Validate domain availability.

Acceptance: Result stored.

08.03 DNS/MX — Check mail-exchange infrastructure.

Acceptance: Result stored.

08.04 Disposable Detection — Detect disposable domains.

Acceptance: Disposable status blocks campaign eligibility.

08.05 Role Detection — Detect role mailboxes such as info/support/sales.

Acceptance: Role status is separately classified.

08.06 Verification Result — Return VALID/INVALID/RISKY/UNKNOWN/DISPOSABLE/ROLE.

Acceptance: Lead campaign eligibility uses status.

M09 — AI Lead Qualification

09.01 Company Analysis — Analyze company, website, industry, size, services, technology, and research.

Acceptance: Structured AI result is stored.

09.02 Lead Score — Score industry fit, company fit, role fit, pain signal, and offer fit.

Acceptance: Overall score is deterministic from configured components.

09.03 Qualification Category — Classify HOT/WARM/COLD/UNQUALIFIED.

Acceptance: Category is visible/filterable.

09.04 Offer Recommendation — Recommend an approved service based on fit.

Acceptance: No invented service claims.

M10 — Lead Management

10.01 Lead Table — Show name, company, title, email, score, status, campaign, last activity.

Acceptance: Search/sort/filter supported.

10.02 Filtering — Filter by score, industry, location, size, email status, lead status, campaign.

Acceptance: Filters are combinable.

10.03 Bulk Actions — Verify, add to campaign, tag, status-change, delete selected leads.

Acceptance: Actions require validation and audit logging.

M11 — Email Account Management

11.01 Add Mailbox — Configure SMTP/IMAP account details.

Acceptance: Credentials encrypted at rest.

11.02 SMTP Test — Test connectivity, auth, TLS, and sending capability.

Acceptance: Clear success/failure result.

11.03 IMAP Test — Test inbox access.

Acceptance: Clear success/failure result.

11.04 Email Health — Show sending, bounce, reply, unsubscribe, and configured volume metrics.

Acceptance: Health state is explainable.

M12 — Email Composer

12.01 Create Email — Compose subject/body and use supported merge variables.

Acceptance: Draft saves correctly.

12.02 Preview — Render personalized variables before send.

Acceptance: Preview reflects selected lead.

M13 — AI Personalization

13.01 Generate Email — Use ICP, research, lead, offer, and campaign context to generate personalized copy.

Acceptance: Output is structured and editable.

13.02 Personalization Rules — Require factual claims, concise copy, approved offers, and no fabricated facts.

Acceptance: AI output passes validation/review rules.

13.03 Preview/Approval — Compare original and AI version and approve/edit.

Acceptance: Only approved content can be sent where approval is configured.

M14 — Campaign Management

14.01 Create Campaign — Create name, description, target ICP, and sender account.

Acceptance: Campaign is saved.

14.02 Select Leads — Add selected leads or ICP segments.

Acceptance: Campaign membership is traceable.

14.03 Sequence — Create multi-step emails and delays.

Acceptance: Sequence is versioned and validated.

M15 — Workflow Automation

15.01 Delays — Wait configurable periods between steps.

Acceptance: Timezone-aware scheduling.

15.02 Business Hours — Restrict sending to configured windows.

Acceptance: Scheduler honors timezone.

15.03 Reply Condition — Stop sequence on reply.

Acceptance: No subsequent automated step is scheduled.

15.04 Bounce Condition — Stop and mark invalid on hard bounce.

Acceptance: Suppression/invalid status applied.

15.05 Unsubscribe Condition — Stop and suppress on unsubscribe.

Acceptance: Global suppression prevents future campaigns.

M16 — Email Sending Engine

16.01 Queue — Place eligible sends into BullMQ.

Acceptance: Queue state is observable.

16.02 Scheduler — Respect send time, timezone, business hours, and daily limits.

Acceptance: No send occurs outside policy.

16.03 Sending Worker — Send via configured SMTP account.

Acceptance: Message and event are persisted.

16.04 Retry — Retry transient failures with bounded backoff.

Acceptance: Permanent failures are not endlessly retried.

M17 — Deliverability

17.01 Authentication Checklist — Check/configure guidance for SPF, DKIM, DMARC.

Acceptance: UI clearly distinguishes configuration checks from guaranteed outcomes.

17.02 Bounce Monitoring — Track hard/soft/unknown bounces.

Acceptance: Hard bounces become ineligible.

17.03 Suppression — Maintain global suppression for unsubscribes and invalid addresses.

Acceptance: All campaigns enforce suppression.

17.04 Sending Controls — Configure daily limits, business hours, delays, and campaign volume.

Acceptance: Controls are enforced server-side.

17.05 Account Health — Show healthy/warning/critical state based on observed signals/configuration.

Acceptance: State has explanatory reasons.

M18 — Reply Management

18.01 IMAP Monitoring — Poll/monitor inboxes for new messages.

Acceptance: New mail creates events/threads.

18.02 Reply Classification — Classify interested, not interested, question, meeting, unsubscribe, wrong person, OOO, other.

Acceptance: Classification is stored with confidence.

18.03 Priority — Prioritize replies by sales intent.

Acceptance: High-priority replies appear prominently.

18.04 Reply Dashboard — Show new, interested, needs response, meetings, unsubscribes.

Acceptance: Filters and actions available.

M19 — AI Reply Assistant

19.01 Generate Reply — Generate a draft using conversation, research, offer, and history.

Acceptance: Draft is editable.

19.02 Suggested Responses — Offer concise/professional/friendly variants.

Acceptance: User chooses/edits before send where approval is enabled.

M20 — CRM

20.01 Pipeline — Stages: New, Qualified, Contacted, Replied, Interested, Meeting, Proposal, Won, Lost.

Acceptance: Stage changes tracked.

20.02 Deal — Store deal name, company, contact, value, stage, probability, expected close.

Acceptance: Deal linked to lead/company.

20.03 Activities — Track emails, replies, calls, meetings, notes, and status changes.

Acceptance: Timeline is chronological.

M21 — Sequence Optimization

21.01 Metrics — Track sent, delivered, bounced, replies, positive replies, meetings, conversions.

Acceptance: Metrics reconcile with events.

21.02 Comparison — Compare campaigns/sequences.

Acceptance: Date and segment filters supported.

21.03 AI Analysis — Analyze subject, opening, CTA, offer, industry, title, timing, and sequence structure.

Acceptance: Recommendations cite available data.

21.04 Recommendations — Suggest measurable changes.

Acceptance: Recommendations can be reviewed, not silently applied.

M22 — A/B Testing

22.01 Variants — Create A/B content variants.

Acceptance: Variants are versioned.

22.02 Split — Assign eligible leads to variants.

Acceptance: Assignment remains stable for the test.

22.03 Compare — Compare reply, positive-reply, and meeting rates.

Acceptance: Results include sample sizes.

M23 — Reporting & Analytics

23.01 Executive Dashboard — Show lead-to-revenue funnel.

Acceptance: Date range supported.

23.02 Funnel — Leads → Qualified → Contacted → Replied → Interested → Meeting → Proposal → Won.

Acceptance: Stage counts are consistent with CRM.

23.03 Exportable Reports — Export operational metrics where required.

Acceptance: Exports respect access controls.

M24 — Tasks & Follow-ups

24.01 Tasks — Create follow-up tasks linked to lead/deal.

Acceptance: Task has due date, priority, status.

24.02 Reminders — Show overdue/upcoming tasks.

Acceptance: Dashboard surfaces critical tasks.

M25 — Tags & Segments

25.01 Tags — Create and assign tags.

Acceptance: Tags are searchable.

25.02 Segments — Combine tags/lead attributes into reusable segments.

Acceptance: Segments can feed campaign selection.

M26 — Import / Export

26.01 Import — Import CSV/XLSX with field mapping.

Acceptance: Validate rows and provide error report.

26.02 Export — Export selected data to CSV/XLSX.

Acceptance: Exports are access-controlled.

M27 — Global Suppression & Compliance

27.01 Suppression List — Store email, reason, date, source.

Acceptance: Suppression is global.

27.02 Consent/Legal Controls — Provide configurable compliance controls and unsubscribe handling appropriate to target jurisdictions.

Acceptance: System should not bypass applicable laws or provider policies.

M28 — Settings

28.01 General — Agency name, timezone, default sending hours.

Acceptance: Settings persist.

28.02 Email — Mailbox configuration and authentication status.

Acceptance: Credentials protected.

28.03 AI — OpenAI key/model/generation controls.

Acceptance: Secrets protected; model settings validated.

28.04 Campaign — Default delays, limits, and sending policies.

Acceptance: Server-side enforcement.

M29 — System Monitoring

29.01 Health — Monitor web, DB, Redis, email, AI, and workers.

Acceptance: Health endpoint/status page.

29.02 Logs — Centralize structured application/worker errors.

Acceptance: Logs include correlation IDs where practical.

M30 — Security

30.01 Secrets — Encrypt mailbox/API credentials at rest.

Acceptance: Encryption key is externally managed.

30.02 Auth Security — Secure sessions, password hashing, rate limiting.

Acceptance: Security tests pass.

30.03 Validation — Validate all API inputs and outputs.

Acceptance: Malformed requests rejected.

30.04 Audit — Record sensitive actions.

Acceptance: Audit trail is immutable/controlled.

M31 — Background Workers

31.01 Discovery Worker — Run discovery jobs asynchronously.

Acceptance: Job progress persisted.

31.02 Enrichment Worker — Process company research.

Acceptance: Retries bounded.

31.03 Verification Worker — Process verification queue.

Acceptance: Results persisted.

31.04 AI Worker — Run AI research/scoring/personalization jobs.

Acceptance: Token/cost metrics tracked.

31.05 Email Worker — Send scheduled mail.

Acceptance: Idempotent sending.

31.06 IMAP Worker — Process incoming messages.

Acceptance: Thread matching reliable.

31.07 Analytics Worker — Aggregate metrics if needed.

Acceptance: Aggregates reconcile with raw events.

M32 — Audit Log

32.01 Activity Audit — Record campaign creation, lead changes, sends, bounces, replies, AI generation, settings changes.

Acceptance: Actor/time/action/resource stored.

M33 — Backups

33.01 Database Backup — Automate PostgreSQL backups.

Acceptance: Restore procedure documented/tested.

33.02 Recovery — Document recovery and disaster procedure.

Acceptance: Recovery test completed.

M34 — Production Deployment

34.01 Docker Deployment — Deploy web, PostgreSQL, Redis, and workers.

Acceptance: Services restart safely.

34.02 Reverse Proxy/TLS — Serve application over HTTPS.

Acceptance: TLS valid and renewed.

34.03 Production Configuration — Separate secrets/config from source code.

Acceptance: No secrets committed to repository.

9. Technical Architecture

Frontend: Next.js + TypeScript + TailwindCSS + shadcn/ui. Application/API: Next.js server-side APIs/actions with strict validation. Database: PostgreSQL + Prisma. Queues: Redis + BullMQ. AI: OpenAI API through a centralized AI service. Email: SMTP for outbound and IMAP for inbound. Workers: Discovery, enrichment, verification, AI, email, IMAP, analytics. Deployment: Docker Compose on Hostinger VPS behind HTTPS reverse proxy.

10. Core Data Model

- User
- Company
- Contact
- Lead
- LeadSource
- LeadActivity
- CompanyResearch
- TechnologySignal
- EmailAccount
- EmailMessage
- EmailThread
- Campaign
- CampaignStep
- CampaignLead
- EmailEvent
- Suppression
- Unsubscribe
- Pipeline

- Deal
- Task
- Note
- AIResearch
- AIScore
- AIMessage
- SearchJob
- SearchResult
- SystemLog
- AuditLog

11. Background Job Architecture

User action → API creates job → BullMQ queue → worker processes job → result/event persisted → UI receives/polls progress. All long-running operations must be asynchronous and idempotent where possible.

12. Email Sending Safety & Reliability Rules

- Only send to campaign-eligible contacts.
- Enforce global suppression before every send.
- Stop sequences after replies where configured.
- Stop after hard bounce.
- Respect configured daily limits, business hours, timezones, and delays.
- Use bounded retries and idempotency to prevent duplicate sends.
- Maintain audit/event records for every send attempt and outcome.

13. AI Guardrails

- AI must not invent company facts, customer relationships, achievements, technology usage, or personal details.
- AI outputs must be structured where consumed by application logic.
- Personalization should use source-backed research.
- Users should be able to review/edit AI-generated outreach and reply drafts.
- AI classification should retain confidence and fallback to human review for ambiguous replies.
- OpenAI usage should be monitored for cost and failures.

14. Security & Privacy Requirements

- Encrypt SMTP/IMAP/API credentials at rest.
- Never commit secrets to Git.
- Use secure password hashing and session handling.
- Validate and sanitize all user-controlled input.
- Apply server-side authorization to every protected operation.
- Use rate limiting on authentication and sensitive endpoints.
- Maintain audit logs for security-sensitive actions.
- Provide data deletion/retention mechanisms appropriate to the product and applicable requirements.

15. MVP Definition

The first usable release should prioritize the user's existing-lead workflow:

- Authentication
- Lead/company/contact CRUD
- CSV/XLSX import
- Email verification
- Mailbox SMTP/IMAP connection
- Email composer
- Campaigns and sequences
- Scheduler + BullMQ
- Sending worker
- Reply detection
- Bounce/unsubscribe handling
- AI personalization
- Basic CRM pipeline
- Basic analytics

16. V2 Definition

- ICP Builder
- Lead discovery engine
- Website intelligence
- Contact discovery
- AI qualification
- Advanced reply classification
- AI reply assistant
- A/B testing
- Sequence optimization
- Advanced analytics

17. V3 / Advanced Platform

- Additional permitted discovery connectors.
- Advanced workflow builder.
- Advanced segmentation.
- Multi-user/team roles if required.
- Advanced AI sales intelligence.
- More sophisticated campaign optimization.
- Expanded monitoring and operational controls.

18. Definition of Done

- Feature works through the UI and API.
- Database migrations are included.
- Validation and authorization are implemented.
- Errors are handled and logged.
- Background jobs are retry-safe.
- Unit/integration tests cover critical logic.

- Security-sensitive values are not exposed.
- Documentation is updated.
- Feature is deployed successfully in staging.

19. Project Delivery Strategy

- Phase 0 — Architecture, repository, environment, Docker, database, Redis
- Phase 1 — Authentication and admin
- Phase 2 — CRM and lead management
- Phase 3 — Existing lead import and email verification
- Phase 4 — Email accounts, SMTP/IMAP, composer
- Phase 5 — Campaigns, sequences, scheduler, sending
- Phase 6 — Reply, bounce, unsubscribe, suppression
- Phase 7 — AI personalization and reply intelligence
- Phase 8 — Lead discovery and website intelligence
- Phase 9 — AI qualification, A/B testing, optimization
- Phase 10 — Analytics, security, testing, backups, production hardening

20. Key Product Metrics

- Qualified lead rate
- Email verification pass rate
- Delivery rate
- Hard bounce rate
- Reply rate
- Positive reply rate
- Meeting rate
- Campaign conversion rate
- Lead-to-meeting conversion
- Meeting-to-deal conversion
- Won revenue
- AI personalization acceptance/edit rate
- AI classification accuracy
- Average response handling time

21. Risks & Mitigations

Email reputation/deliverability: Use authenticated domains/mailboxes, controlled sending, suppression, bounce monitoring, and provider-compliant practices.

Lead data quality: Use source metadata, normalization, verification, confidence scores, and user review.

External source changes: Use connector interfaces so discovery sources can be replaced independently.

AI hallucination: Source-backed prompts, structured output, validation, and human approval.

Duplicate sends: Idempotency keys, unique message identifiers, and transactional state changes.

VPS resource limits: Asynchronous queues, worker concurrency controls, caching, indexes, and monitoring.

Compliance risk: Jurisdiction-aware policies, unsubscribe/suppression controls, data minimization, and legal review before production outreach.

22. Product Principle

The platform should not be designed as a generic bulk-email tool. Its primary value is the intelligence loop: discover/ingest → qualify → research → personalize → automate → observe → understand replies → optimize → convert. Lead generation, CRM, email infrastructure, and analytics exist to support this loop.

23. Recommended First Engineering Sprint

- Create monorepo and development conventions.
- Initialize Next.js + TypeScript + Tailwind + shadcn/ui.
- Provision PostgreSQL and Redis locally with Docker.
- Set up Prisma schema and first migration.
- Implement authentication and protected dashboard.
- Implement User, Company, Contact, Lead, Activity models.
- Build lead CRUD UI and API.
- Add CSV import with field mapping.
- Add structured logging and error handling.
- Set up CI checks and basic tests.

END OF PRD — Version 1.0

M35 — Email Warmup & Mailbox Reputation

Email Warmup is a first-class deliverability module designed to support responsible mailbox health, gradual sending, authentication checks, and monitoring. It must not promise inbox placement or use deceptive/fake engagement.

35.01 Warmup Configuration — Select mailbox, start date, daily warmup limit, growth schedule, maximum volume, sending hours, and timezone.

Acceptance: Settings persist and are enforced by the server.

35.02 Warmup Scheduler — Create scheduled warmup jobs and process them through a dedicated queue/worker.

Acceptance: Jobs respect mailbox limits, time windows, and pause state.

35.03 Warmup Status — Show active/paused status, warmup day, current volume, target volume, and health state.

Acceptance: Status reflects persisted job/mailbox data.

35.04 Authentication Checks — Provide SPF, DKIM, and DMARC configuration/status guidance.

Acceptance: Authentication state is visible and clearly separated from inbox-placement claims.

35.05 Reputation/Health Monitoring — Track bounces, complaints when available, unsubscribes, delivery failures, sending volume, and mailbox errors.

Acceptance: Health state is explainable as GOOD/WARNING/CRITICAL.

35.06 Safety Controls — Pause or reduce warmup/campaign sending based on configured safety thresholds and authentication/mailbox failures.

Acceptance: Safety controls apply before send execution.

35.07 Warmup/Campaign Separation — Maintain separate warmup and campaign queues under a shared mailbox-level sending policy.

Acceptance: Combined sending cannot silently exceed configured limits.

M35 Implementation Principle

Warmup is not a magic inbox-placement mechanism. The platform should prioritize authentication, gradual volume changes, bounce/complaint monitoring, suppression, mailbox health, and responsible sending behavior. It must not manufacture fake engagement or misleading replies.